EQSANS Chopper Calibration Tools

User Guide and Phase Offset Determination

EQSANS Instrument Team
Spallation Neutron Source, ORNL

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Abstract

This guide describes the Python-based tools for EQSANS chopper calibration and phase verification. The two primary tools are `find_edges_ui.py` for extracting phase offsets from experimental monitor spectra, and `eqsans_chopper_sim.py` for simulating multi-chopper neutron transmission and verifying phase configurations. This document provides step-by-step workflows, explains the underlying physics, and includes practical examples for both 60 Hz and 30 Hz operating modes.

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1 Introduction

The EQSANS instrument at SNS uses a system of six disk choppers (1a, 1b, 2a, 2b, 3a, 3b) to define the wavelength band transmitted to the sample and detector. Precise timing (phase) of each chopper is critical for:

- Selecting the desired wavelength range
- Minimizing frame overlap from adjacent source pulses
- Enabling frame-skip mode for extended wavelength coverage at 30 Hz

The calibration workflow involves two complementary approaches:

1. **Experimental measurement**: Using `find_edges_ui.py` to analyze monitor spectra and extract the actual chopper phase offsets from measured TOF edges.
2. **Simulation verification**: Using `eqsans_chopper_sim.py` to predict the wavelength bands for given phase settings and verify consistency with the control system (hexaSub).

2 Physical Background

2.1 Chopper Timing and TOF

Neutrons travel from the moderator to each chopper with time-of-flight (TOF) determined by their wavelength λ and the chopper distance L:

$$\mathrm{TOF} = \frac{L \cdot \lambda}{h/m_n} = \frac{L \cdot \lambda}{3956\,\mathrm{m\,Å/ms}} \tag{1}$$

Each chopper has an opening gap characterized by:

- **Opening angle** θ: The angular width of the gap (in degrees)
- **OGC (Opening Gap Center)**: The time when the center of the gap aligns with the beam
- **Opening time**: $t_{\mathrm{open}} = (\theta/360) \times T_{\mathrm{period}}$

2.2 Chopper Configuration

Table 1 lists the EQSANS chopper parameters.

Table 1: EQSANS Chopper Parameters

Chopper	Distance (m)	Opening (°)	Opening at 60 Hz (μs)
1a	5.6601	129.605	6000
1b	5.6758	129.605	6000
2a	7.7601	179.989	8333
2b	7.7758	179.989	8333
3a	9.4978	230.010	10648
3b	9.5078	230.007	10648

2.3 60 Hz vs 30 Hz Operation

60 Hz mode: Choppers rotate at 60 Hz (period = 16,667 μs), synchronized with the SNS source frequency. Each chopper opening window accepts neutrons from a single source pulse. Figure 1 shows a distance–time diagram for 60 Hz operation with a 2.5 Å starting wavelength.

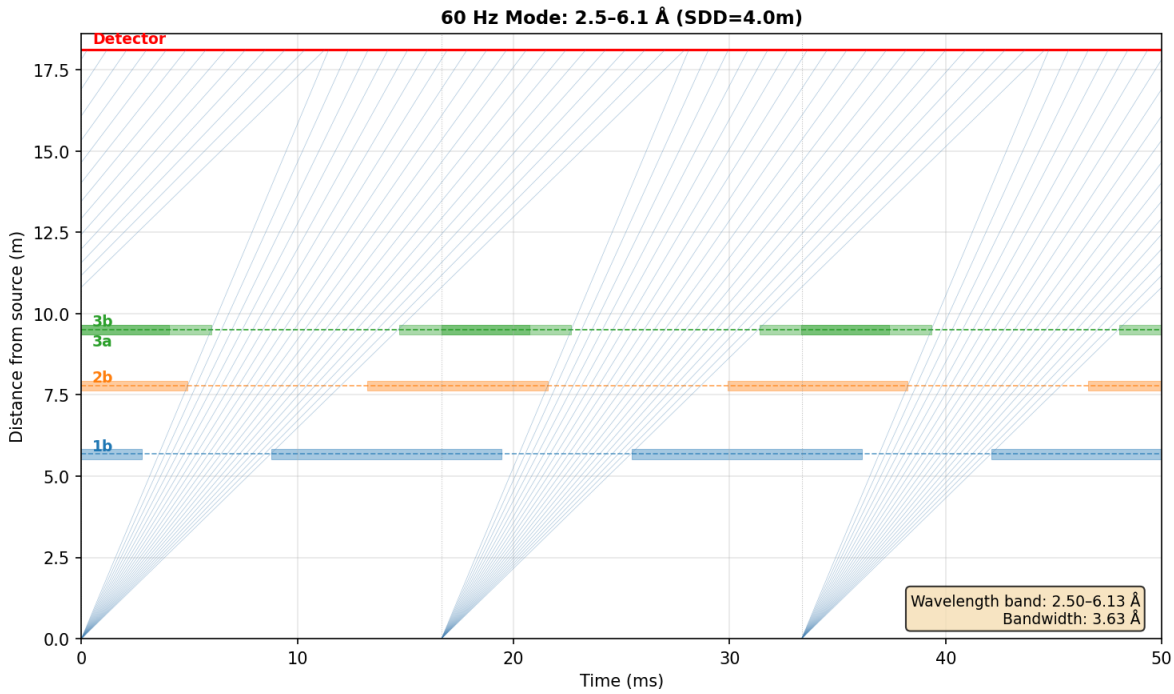

Figure 1: Distance–time diagram for 60 Hz mode (2.5–6.1 Å band). Shaded rectangles indicate chopper closed periods. Blue trajectories show neutrons that successfully pass through all four choppers (1b, 2b, 3a, 3b). The choppers define a single wavelength band per source pulse.

30 Hz mode (frame-skip): Choppers rotate at 30 Hz (period = 33,333 μs), but the source still operates at 60 Hz. This creates two wavelength bands per chopper cycle:

- **Band 1**: Neutrons from the "current" source pulse
- **Band 2**: Neutrons from the "previous" source pulse (16.67 ms earlier)

Different choppers align to different pulses in frame-skip mode, enabling extended wavelength coverage with a gap between bands. Figure 2 illustrates the frame-skip configuration.

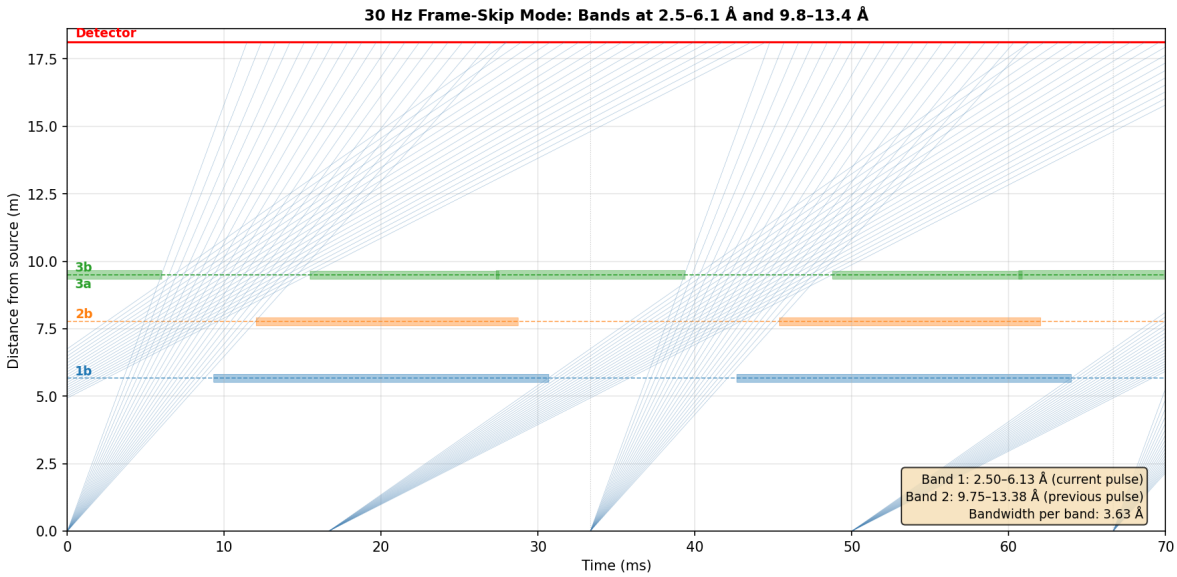

Figure 2: Distance–time diagram for 30 Hz frame-skip mode. Two distinct wavelength bands are visible: Band 1 (2.5–6.1 Å) from the current pulse, and Band 2 (9.8–13.4 Å) from the previous pulse. The choppers 1b and 3a align to the previous pulse while 2b and 3b align to the current pulse, creating the characteristic dual-band pattern.

2.4 Monochromatic Mode

For applications requiring a narrow wavelength band centered on a specific wavelength, all six choppers can be used in monochromatic mode. In this mode, hexaSub calculates phases such that:

- **"a" choppers** (1a, 2a, 3a): Opening edge aligned to λ_{min}
- **"b" choppers** (1b, 2b, 3b): Closing edge aligned to λ_{max}

Figure 3 shows a configuration for 10 Å with ±10% spread (9.5–10.5 Å). Table 2 lists the hexaSub-calculated OGC phases.

Table 2: hexaSub OGC Phases for Monochromatic Mode (10 Å ± 10%, SDD=4 m)

Chopper	Aligned to	Wavelength (Å)	OGC Phase (μs)
1a	λ_{min} (opening)	9.50	14873.50
1b	λ_{max} (closing)	10.50	10462.74
2a	λ_{min} (opening)	9.50	4273.76
2b	λ_{max} (closing)	10.50	14639.37
3a	λ_{min} (opening)	9.50	9525.20
3b	λ_{max} (closing)	10.50	1143.60

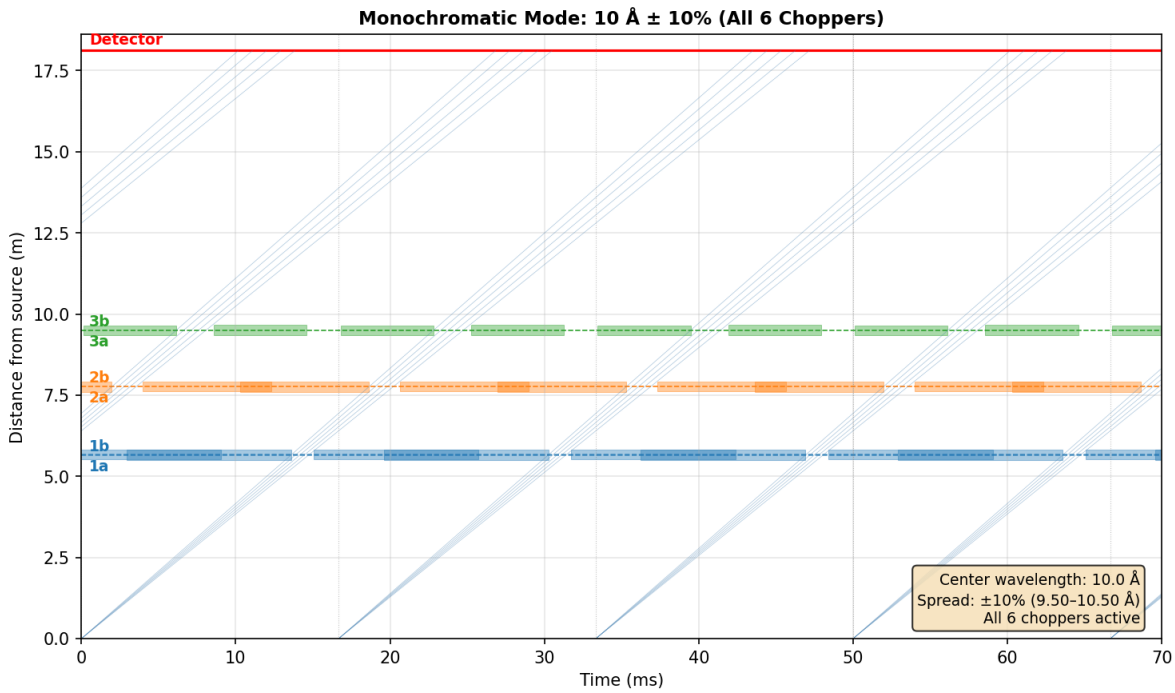

Figure 3: Distance–time diagram for monochromatic mode: 10 Å ± 10% (9.5–10.5 Å). All six choppers are active with phases calculated by hexaSub. The "a" choppers define the lower wavelength cutoff while "b" choppers define the upper cutoff, creating a precisely bounded wavelength band.

3 Tool 1: Interactive Edge Finder (`find_edges_ui.py`)

3.1 Purpose

The `find_edges_ui.py` tool analyzes monitor spectra to:

- Identify the TOF positions of chopper blade edges (rising/falling transitions)
- Fit edges using an error function (erf) model
- Calculate the phase offset between measured edges and expected chopper timing

3.2 Starting the Tool

```
# Load spectrum from ASCII file (60 Hz mode)
python find_edges_ui.py 176475_monitor.dat

# Load spectrum in 30 Hz mode
python find_edges_ui.py --30hz 176550_monitor.dat

# Start without file (load via Phase Calculator)
python find_edges_ui.py
```

3.3 User Interface Overview

The interface consists of two windows (Figure 4):

1. **Main Plot Window**: Displays the TOF spectrum with interactive edge marking
2. **Phase Calculator Window**: Computes OGC and phase offsets from edge positions

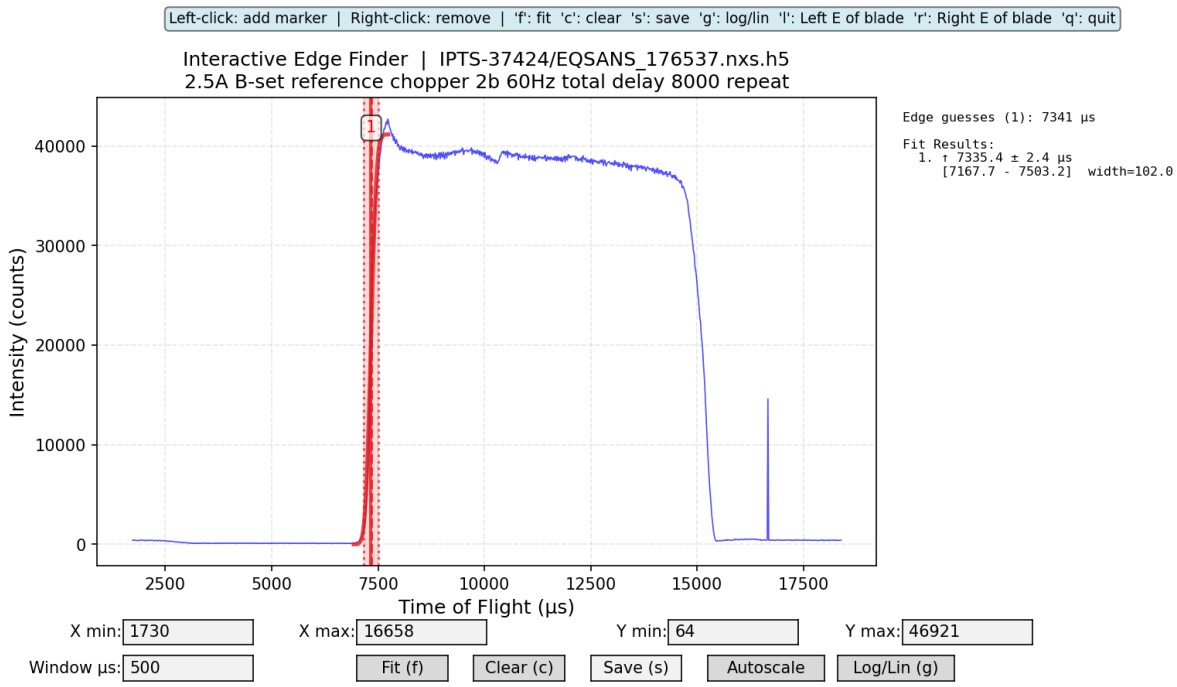

Figure 4: Interactive Edge Finder showing a 60 Hz monitor spectrum with a fitted rising edge. The fit result panel shows the edge position ($7335.4 \pm 2.4\,\mu$s), fit range, and transition width.

3.4 Keyboard Shortcuts and Controls

Table 3: Interactive Controls

Action	Control
Add edge marker	Left-click on plot
Remove nearest marker	Right-click on plot
Fit all marked edges	Press `f` or click "Fit"
Clear all markers	Press `c` or click "Clear"
Save results (data + image)	Press `s` or click "Save"
Toggle log/linear Y-scale	Press `g` or click "Log/Lin"
Set left blade edge to cursor	Press `l`
Set right blade edge to cursor	Press `r`
Quit	Press `q`

3.5 Step-by-Step Workflow: Extracting Phase Offset

3.5.1 Step 1: Load the Monitor Spectrum

Option A — From ASCII file:

```
python find_edges_ui.py edge_refinement/176505_monitor.dat
```

Option B — From NeXus via Phase Calculator:

1. Start the tool without a file: `python find_edges_ui.py`
2. In the Phase Calculator window, enter the IPTS number and Run number
3. Click "Load" to fetch the spectrum from the NeXus file

3.5.2 Step 2: Mark Edge Positions

1. Identify the rising and/or falling edges in the spectrum (transitions where intensity changes sharply)
2. Left-click at each edge location to place a numbered marker
3. Use right-click to remove incorrectly placed markers

3.5.3 Step 3: Fit the Edges

Press `f` (or click "Fit") to fit all marked edges. The tool:

1. Automatically detects whether each edge is rising or falling
2. Fits an error function (erf) model to determine the precise edge position
3. Displays the fit results: position ± uncertainty, width, and boundary region

The erf model used for fitting is:

$$I(t) = \frac{I_R - I_L}{2}\left[1 + \mathrm{erf}\left(\frac{t - t_0}{\sqrt{2}\sigma}\right)\right] + I_L \tag{2}$$

where t_0 is the edge position, σ is the transition width, and I_L, I_R are the left/right intensity levels.

3.5.4 Step 4: Capture Edge Values for Phase Calculation

1. In the Phase Calculator, select the chopper being calibrated (e.g., 1b, 2a, etc.)
2. Position the mouse cursor at the fitted edge position
3. Press `l` to set as "Left E of blade" or `r` for "Right E of blade"
4. Alternatively, type the TOF value directly into the input field

3.5.5 Step 5: Read the Phase Offset

The Phase Calculator computes and displays:

- **OGC**: The Opening Gap Center time calculated from the edge position
- **Phase offset**: The difference between the current offset setting and the calculated OGC

The OGC is calculated as:

$$\mathrm{OGC} = \frac{L \cdot t_{\mathrm{edge}}}{k} \pm \frac{\theta}{360} \cdot \frac{T_{\mathrm{period}}}{2} \tag{3}$$

where the sign depends on whether it's the left (opening, +) or right (closing, −) edge.

3.5.6 Step 6: Save Results

Press `s` to save:

- ASCII file with edge fit results (position, uncertainty, width, amplitude)
- PNG image of the spectrum with fitted edges

3.6 Example: 30 Hz Frame-Skip Mode

Figure 5 shows edge detection in 30 Hz mode, where the frame period extends to 33,333 μs.

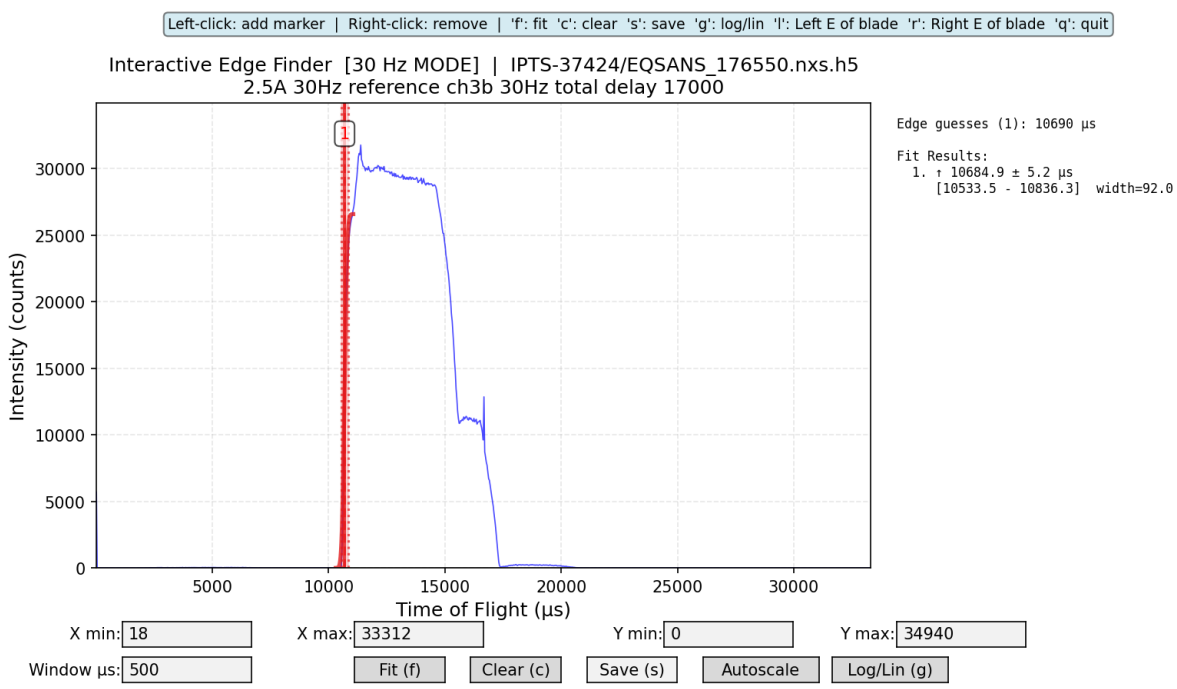

Figure 5: Edge detection in 30 Hz frame-skip mode. The extended TOF range (0–33,333 μs) captures the full chopper period. The fitted edge at 10,685 μs corresponds to a 2.5 Å reference measurement.

4 Tool 2: Multi-Chopper Simulator (`eqsans_chopper_sim.py`)

4.1 Purpose

The `eqsans_chopper_sim.py` tool provides:

- Visual simulation of neutron trajectories through multiple choppers
- Prediction of wavelength bands for given phase settings
- Verification that hexaSub-calculated phases produce expected results
- Distance–time diagrams showing chopper open/closed states

4.2 Starting the Tool

```
python eqsans_chopper_sim.py
```

4.3 User Interface Overview

The simulator GUI consists of:

- **Control Panel (left)**: Frequency selection, chopper enable/disable, OGC and mechanical offset inputs, hexaSub checkbox
- **Distance–Time Diagram (top right)**: Shows chopper positions, open/closed windows, and neutron trajectories
- **Detector TOF Spectrum (bottom right)**: Histogram of neutrons arriving at the detector

4.4 Control Panel Elements

Table 4: Simulator Controls

Control	Description
Detector distance	Physical distance from sample to detector (meters)
Frequency (60/30 Hz)	Chopper rotation frequency; affects period and frame-skip behavior
Chopper checkboxes	Enable/disable individual choppers in the simulation
OGC (μs)	Opening Gap Center phase for each chopper
Mech offset (μs)	Mechanical offset added to OGC (calibration correction)
hexaSub phases	When checked, auto-populates mech offsets for hexaSub compatibility
Calculate	Run the simulation with current settings
Log scale	Toggle logarithmic Y-axis on TOF spectrum

4.5 Understanding the hexaSub Checkbox

The EQSANS chopper control system uses `hexaSub.c` to calculate OGC phases for a requested wavelength. These phases include hardware calibration offsets (∼29,000–30,000 μs) that account for encoder alignment, cable delays, and other instrument-specific factors.

When the **hexaSub phases** checkbox is enabled:

1. The simulator automatically sets mechanical offsets to compensate for hexaSub calibration
2. For 30 Hz mode, additional frame-skip alignment corrections are applied
3. You can directly paste hexaSub OGC values into the OGC fields

Table 5 shows the mechanical offsets applied for 30 Hz mode.

Table 5: Mechanical Offsets with hexaSub Checkbox (30 Hz Mode)

Chopper	Calibration Offset (μs)	Frame-skip Align (μs)	Net Mech Offset (μs)
1b	−29908.92	0	−29908.92
2b	−29610.80	0	−29610.80
3a	−29452.12	+33333.33	+3881.21
3b	−29131.20	+33333.33	+4202.13

4.6 Step-by-Step Workflow: Verifying hexaSub Phases

4.6.1 Step 1: Configure the Frequency

Select **30 Hz** or **60 Hz** to match the chopper operating mode.

4.6.2 Step 2: Enable the Appropriate Choppers

For 30 Hz frame-skip mode, enable choppers **1b, 2b, 3a, 3b** (the "b-set" plus 3a/3b).

4.6.3 Step 3: Enable hexaSub Phases

Check the **hexaSub phases** checkbox. This automatically sets the mechanical offsets.

4.6.4 Step 4: Enter hexaSub OGC Values

Enter the OGC values calculated by hexaSub for your wavelength setting. For example, for 2.5 Å at 30 Hz with SDD=4 m:

Chopper	hexaSub OGC (μs)
1b	33233.61
2b	33318.81
3a	921.45
3b	12454.81

4.6.5 Step 5: Run the Simulation

Click **Calculate**. The simulator will display:

- Distance–time diagram with chopper windows and neutron trajectories
- Detector TOF spectrum showing the wavelength bands that pass

4.6.6 Step 6: Verify the Results

For the 2.5 Å @ 30 Hz example, you should see two wavelength bands:

- **Band 1**: ∼2.5–6.1 Å (from current source pulse)
- **Band 2**: ∼9.8–13.4 Å (from previous source pulse)

The status bar reports the number of pulse groups that passed and the peak TOF.

4.7 Understanding the Distance–Time Diagram

The diagram shows:

- **Horizontal axis**: Time (ms) from source pulse
- **Vertical axis**: Distance from source (m)
- **Shaded rectangles**: Chopper closed periods (color-coded by chopper)
- **Horizontal dashed lines**: Chopper positions with labels showing name, distance, and effective OGC
- **Blue trajectory lines**: Sampled neutron paths that successfully pass all choppers
- **Top line**: Detector position

Neutrons that intersect a shaded (closed) region are blocked. Only trajectories passing through all open windows reach the detector.

4.8 Relationship to hexaSub.c

The `hexaSub.c` code in the EPICS IOC calculates chopper phases using the `calc6Phases()` function. This section provides detailed explanations of phase calculation for all three operating modes.

4.8.1 Common Elements

All modes share these fundamental calculations:

- **TOF calculation**: Time-of-flight from source to chopper position using $\text{TOF} = d/(3956 \times 10^3) \times \lambda$ where d is in mm and λ in Å
- **Centering adjustment**: After aligning an edge to a wavelength, the phase is adjusted by $\pm\theta_{\text{opening}} \times t_{\text{half-angle}}$ to center the OGC (Opening Gap Center) on the opening window
- **Calibration offset**: Hardware-specific offset (∼15,000 μs at 60 Hz, ∼30,000 μs at 30 Hz) added to account for encoder alignment, cable delays, etc.

- **Period wrapping**: Final phase wrapped to one period via `fmod(phase, frame_width)`

The general formula for each chopper is:

$$\mathrm{OGC} = \texttt{fmod}\Big(10^6 \times \big(\mathrm{TOF}_\lambda \pm \theta_{\text{opening}} \times t_{\text{half-angle}}\big) + \mathrm{offset}_{\text{cal}}\,,\ T_{\text{frame}}\Big) \tag{4}$$

4.8.2 60 Hz Mode (Standard Operation)

In 60 Hz mode, hexaSub uses choppers indexed 1–6 in the code. The array index mapping is:

```
index:  1=1b, 2=2b, 3=3a, 4=3b, 5=1a, 6=2a
```

The phase offsets are:

```
phase_offset[] = {0, 14954.46, 14805.4, 14726.06, 14565.69, 15072.89, 14834.04}
                    (1b)       (2b)      (3a)      (3b)      (1a)      (2a)
```

Wavelength-dependent alignment (critical):
The 60 Hz mode has *different* alignment logic depending on whether $\lambda_1 > 13$ Å:

Table 6: 60 Hz Mode Edge Alignment

Condition	1a/1b Edge	Aligned to	2a/2b Edge	Aligned to
$\lambda_1 > 13$ Å	Opening	λ_1	Closing	λ_2
$\lambda_1 \leq 13$ Å	Closing	λ_2	Opening	λ_1

This swap prevents neutron leakage at different wavelength ranges. The choppers 3a and 3b do *not* swap:

- **3a**: Closing edge always aligned to λ_2 (band end)
- **3b**: Opening edge always aligned to λ_1 (band start)

Phase calculation (60 Hz):
For $\lambda_1 \leq 13$ Å (most common):

$$\mathrm{phase}_{1a/b} = \mathrm{TOF}(\lambda_2) - \theta_{\text{open}} \times t_{\text{half}} + \text{offset} \quad \text{(closing edge)} \tag{5}$$

$$\mathrm{phase}_{2a/b} = \mathrm{TOF}(\lambda_1) + \theta_{\text{open}} \times t_{\text{half}} + \text{offset} \quad \text{(opening edge)} \tag{6}$$

$$\mathrm{phase}_{3a} = \mathrm{TOF}(\lambda_2) - \theta_{\text{open}} \times t_{\text{half}} + \text{offset} \quad \text{(closing edge)} \tag{7}$$

$$\mathrm{phase}_{3b} = \mathrm{TOF}(\lambda_1) + \theta_{\text{open}} \times t_{\text{half}} + \text{offset} \quad \text{(opening edge)} \tag{8}$$

4.8.3 30 Hz Frame-Skip Mode

In 30 Hz mode, choppers rotate at half speed, allowing two source pulses to contribute per chopper period. This creates *two* wavelength bands. Only choppers 1b, 2b, 3a, 3b are used (indices 1, 2, 3, 4).

Phase offsets (30 Hz):
The 30 Hz phase offsets are exactly *double* the 60 Hz values:

```
phase_offset[] = {0, 29908.92, 29610.80, 29452.12, 29131.38, 30145.78, 29668.08}
                    (1b)       (2b)       (3a)      (3b)      (1a)      (2a)
```

This is 2× the 60 Hz offsets because the chopper period doubles from 16,666.67 μs to 33,333.33 μs.

Wavelength bands:

hexaSub computes four wavelengths for frame-skip mode:

- λ_1: Band 1 start (user-requested wavelength)
- $\lambda_2 = \lambda_1 + \Delta\lambda_{60\text{Hz}}$: Band 1 end
- $\lambda_3 = \lambda_2 + \Delta\lambda_{60\text{Hz}}$: Band 2 start
- $\lambda_4 = \lambda_3 + \Delta\lambda_{60\text{Hz}}$: Band 2 end

Pulse alignment (critical):

Different choppers align to *different* source pulses:

Table 7: 30 Hz Mode Chopper Alignment

Chopper	Edge Type	Wavelength	Source Pulse
1b (index 1)	Opening	λ_3	Previous ($-1/60$ s)
2b (index 2)	Closing	λ_2	Current
3a (index 3)	Closing	λ_4	Previous ($-1/60$ s)
3b (index 4)	Opening	λ_1	Current

The "previous pulse" alignment means the TOF calculation *always* includes subtraction of 1/60 s (one source period) for choppers 1b and 3a:

```
// For choppers 1b (index 1) and 1a (index 5):
phase = tof - 1.0/60.0;  // Always subtract, regardless of tof value

// For chopper 3a (index 3):
phase = tof - 1.0/60.0;  // Always subtract
```

This creates the frame-skip behavior where Band 2 neutrons (longer wavelengths) from the *previous* source pulse arrive at the detector in the same time window as Band 1 neutrons from the *current* pulse.

Simulator frame-skip corrections:

When using hexaSub phases in the simulator (checkbox enabled), frame-skip alignment corrections are applied only for choppers 3a and 3b:

- 3a: +33333.33 μs (full 30 Hz period)
- 3b: +33333.33 μs (full 30 Hz period)

These corrections compensate for the period-wrapping in hexaSub's `fmod()` operation. Choppers 1b and 2b require no additional alignment correction because hexaSub always subtracts 1/60 s for 1b, resulting in phase values that already account for pulse alignment.

4.8.4 Monochromatic Mode

Monochromatic mode uses all six choppers to produce a narrow wavelength band (e.g., 10 Å ±10%). This mode operates at 60 Hz and uses the 60 Hz phase offsets.

Phase offsets (same as 60 Hz):

```
phase_offset[] = {0, 14954.46, 14805.4, 14726.06, 14565.69, 15072.89, 14834.04}
                    (1b)       (2b)      (3a)      (3b)      (1a)      (2a)
```

Simple alignment rule:
Unlike the 60 Hz standard mode, monochromatic mode has *no* wavelength-dependent switching:

- **All "a" choppers** (1a, 2a, 3a): Opening edge aligned to λ_{min}
- **All "b" choppers** (1b, 2b, 3b): Closing edge aligned to λ_{max}

Table 8: Monochromatic Mode Edge Alignment

Chopper	Edge Type	Aligned to
1b (index 1)	Closing	λ_{max}
2b (index 2)	Closing	λ_{max}
3a (index 3)	Opening	λ_{min}
3b (index 4)	Closing	λ_{max}
1a (index 5)	Opening	λ_{min}
2a (index 6)	Opening	λ_{min}

Phase calculation (monochromatic):

$$\text{phase}_{na} = \text{TOF}(\lambda_{\text{min}}) + \theta_{\text{open}} \times t_{\text{half}} + \text{offset} \quad \text{(opening edge)} \tag{9}$$

$$\text{phase}_{nb} = \text{TOF}(\lambda_{\text{max}}) - \theta_{\text{open}} \times t_{\text{half}} + \text{offset} \quad \text{(closing edge)} \tag{10}$$

The code comments note: "no funny business is required for wavelengths above or below 13 Å like is shown in the regular 60 Hz mode."

Example: 10 Å ±10% (9–11 Å):
Using hexaSub to calculate phases for 10 Å center wavelength with 10% spread:

Chopper	hexaSub OGC (μs)
1a	14873.50
1b	10462.74
2a	4273.76
2b	14639.37
3a	9525.20
3b	1143.60

Table 9: Comparison of hexaSub Operating Modes

Feature	60 Hz Standard	30 Hz Frame-Skip	Monochromatic
Chopper speed	60 Hz	30 Hz	60 Hz
Choppers used	4 (1a/b, 2a/b or 3a/b)	4 (1b, 2b, 3a, 3b)	6 (all)
Wavelength bands	1	2	1 (narrow)
Phase offsets	∼15,000 μs	∼30,000 μs	∼15,000 μs
$\lambda > 13$ Å switch	Yes	No	No
Pulse alignment	Single pulse	Split (prev/curr)	Single pulse

4.8.5 Summary: Mode Comparison

4.8.6 Using hexaSub Phases in the Simulator

The simulator reverses hexaSub's operations to recover physical timing for ray-tracing:

1. **Subtract calibration offset**: Removes the hardware-specific phase offset
2. **Add frame-skip corrections** (30 Hz only): Compensates for period wrapping
3. **Ray-trace**: Simulates neutron paths through all enabled choppers

When the **hexaSub phases** checkbox is enabled, these corrections are applied automatically, allowing direct use of OGC values from hexaSub.

5 Practical Examples

5.1 Example 1: Single Chopper Calibration (60 Hz)

Goal: Determine the phase offset for chopper 1b from a calibration run.

1. Acquire a monitor spectrum with only chopper 1b active
2. Load the spectrum: `python find_edges_ui.py 176479_1b_pd1000_monitor.dat`
3. Identify and mark the rising and falling edges
4. Press `f` to fit both edges
5. In Phase Calculator, select "1b" and capture each edge using `l` and `r`
6. Record the phase offset values for left and right edges
7. The average offset indicates the calibration correction needed

5.2 Example 2: Verifying 30 Hz Frame-Skip Configuration

Goal: Confirm that hexaSub phases for 2.5 Å @ 30 Hz produce the expected bands.

1. Start the simulator: `python eqsans_chopper_sim.py`
2. Select **30 Hz** frequency

3. Enable choppers 1b, 2b, 3a, 3b (disable 1a, 2a)
4. Check **hexaSub phases**
5. Enter OGC values: 1b=33233.61, 2b=33318.81, 3a=921.45, 3b=12454.81
6. Click **Calculate**
7. Verify two bands appear: ∼2.5–6.1 Å and ∼9.8–13.4 Å

5.3 Example 3: Diagnosing Phase Mismatch

Symptom: Measured wavelength band doesn't match expected range.
Diagnostic workflow:

1. Use `find_edges_ui.py` to measure actual edge positions from monitor data
2. Compare measured edges to hexaSub-predicted values
3. In the simulator, enter the *measured* OGC values (with hexaSub checkbox *off*)
4. Observe which wavelengths pass — this shows the actual instrument state
5. The difference between expected and measured OGC indicates the calibration drift

6 Troubleshooting

6.1 No Wavelengths Pass in Simulator

- Check that choppers are enabled (checkboxes)
- Verify OGC values are reasonable (typically 0–33,333 μs for 30 Hz)
- If using hexaSub phases, ensure the checkbox is checked
- Try with only one or two choppers to isolate timing issues

6.2 Edge Fit Fails

- Increase the "Window" parameter to include more data around the edge
- Ensure the marker is placed near the actual transition, not on flat regions
- Check that the spectrum has sufficient statistics (low-count data may not fit well)

6.3 Phase Calculator Shows Unexpected Values

- Verify the correct chopper is selected
- Check that 30 Hz mode is enabled if using 30 Hz data (title shows "[30 Hz MODE]")
- Confirm the "Current offset" field contains the expected value for comparison

7 File Formats

7.1 Monitor Spectrum (.dat)

ASCII format with two columns:

```
# Optional title line starting with #
TOF_us    Intensity
1000.0    523
1010.0    547
...
```

7.2 Edge Fit Results (.dat)

Output from `find_edges_ui.py` save function:

```
# Edge Detection Results (Interactive UI)
# Columns: edge_num edge_type position position_err start end width amplitude
1 rising 7335.4499 2.4172 7167.7388 7503.1609 101.9611 41168.3902
```

8 References

- EQSANS Instrument: https://neutrons.ornl.gov/eqsans
- hexaSub.c source: `bl6chopper_src/hexaSub.c`
- Tool source code: `/EQSANS/shared/script/choppers/`